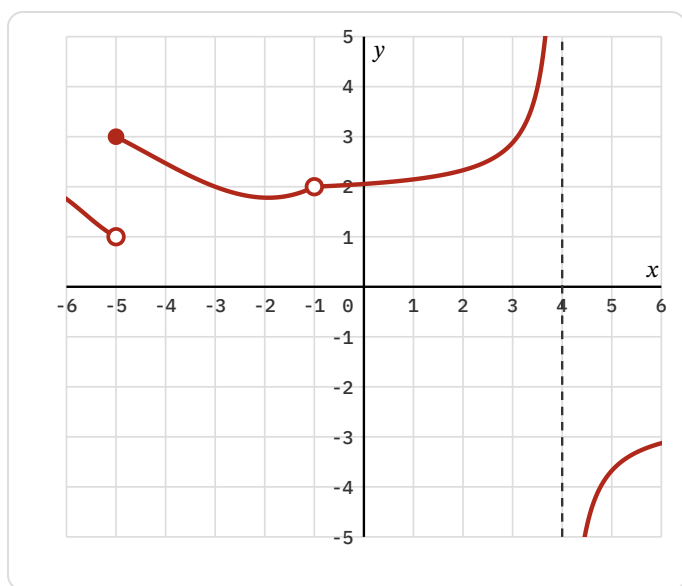


Mock Midterm C

Practice midterm (AI-Free) · Sections 1.1–1.4

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-1)$ _____ (1)
- b.) $\lim_{x \rightarrow -1^-} f(x)$ _____ (1)
- c.) $\lim_{x \rightarrow -1^+} f(x)$ _____ (1)
- d.) $\lim_{x \rightarrow -1} f(x)$ _____ (1)
- e.) $f(4)$ _____ (1)
- f.) $\lim_{x \rightarrow 4^-} f(x)$ _____ (1)
- g.) $\lim_{x \rightarrow 4^+} f(x)$ _____ (1)
- h.) $\lim_{x \rightarrow 4} f(x)$ _____ (1)

2. Let $f(x) = \frac{-2}{(x-3)^2}$. Answer with a number, ∞ , $-\infty$, or DNE.

5 pts

a.) $\lim_{x \rightarrow 3^-} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow \infty} f(x)$ _____ (1)

e.) The vertical asymptote is $x =$ _____ (1)

3. Suppose

4 pts

$$\lim_{x \rightarrow 3} f(x) = -3, \quad \lim_{x \rightarrow 3} g(x) = 4, \quad \lim_{x \rightarrow 3} h(x) = 9$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 3} x^2 f(x)$ _____ (1)

b.) $\lim_{x \rightarrow 3} \frac{f(x) g(x)}{h(x)+4}$ _____ (1)

c.) $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2$ _____ (1)

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{x-36}$ _____ (2)

b.) $\lim_{x \rightarrow -2} \frac{3x^2+4x-4}{x^2+6x+8}$ _____ (2)

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results. 6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-3x+7}{-2x^2-9x}$ _____ (2)

b.) $\lim_{x \rightarrow -\infty} \frac{-x^3}{6x^2+6x}$ _____ (2)

c.) $\lim_{x \rightarrow \infty} \frac{4x^3+6x^2+6x}{-5x^3-8x^2+1}$ _____ (2)

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound.

4 pts

a.) $\lim_{x \rightarrow -\infty} (-4x^5 + 5x^2 + 3x)$ _____ (1)

b.) $\lim_{x \rightarrow -\infty} (-2x^5 - 3x^4 + x)$ _____ (1)

c.) $\lim_{x \rightarrow -\infty} \left(-3 - \frac{8}{x^3}\right)$ _____ (1)

d.) $\lim_{x \rightarrow -\infty} (-e^{-x})$ _____ (1)

7. Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 + 15x + 25}{2x + 5} & x < -\frac{5}{2} \\ -2x - 3 & x \geq -\frac{5}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -5} f(x)$ _____ (1)

c.) $f\left(-\frac{5}{2}\right)$ _____ (1)

d.) $\lim_{x \rightarrow -\frac{5}{2}^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow -\frac{5}{2}^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow -\frac{5}{2}} f(x)$ _____ (1)

8. Find the value of k that makes f continuous at $x = 1$.

4 pts

$$f(x) = \begin{cases} x^2 - x & x < 1 \\ kx - 6 & x \geq 1 \end{cases}$$

a.) $\lim_{x \rightarrow 1^-} f(x)$ _____ (2)

b.) $k =$ _____ (2)

9. Classify each function at the given x -value.

3 pts

a.) $f(x) = \begin{cases} x-2 & x < 2 \\ x-4 & x \geq 2 \end{cases}$ at $x = 2$ (1)

(A) Continuous

(B) Removable discontinuity

(C) Jump discontinuity

(D) Infinite discontinuity

b.) $f(x) = \frac{x^2-16}{x-4}$ at $x = 4$ (1)

(A) Continuous

(B) Removable discontinuity

(C) Jump discontinuity

(D) Infinite discontinuity

c.) $f(x) = \frac{x+2}{x^2+6x+8}$ at $x = -4$ (1)

(A) Continuous

(B) Removable discontinuity

(C) Jump discontinuity

(D) Infinite discontinuity

10. Let $f(x) = x^2 + 6x + 5$, where $f(x)$ represents the number of miles driven after x hours. 7 pts
Find the average rate of change over $[2, 6]$.

a.) $f(2)$ _____ (2)

b.) $f(6)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

- d.) Which sentence correctly states the result **with units**? (1)

(A) The distance driven changes by a total of **14** miles from $x = 2$ to $x = 6$.

(B) Between $x = 2$ and $x = 6$, the distance driven changes at an average rate of **14** hours per mile.

(C) At exactly $x = 6$ hours, the distance driven is changing at a rate of **14** miles per hour.

(D) Between $x = 2$ and $x = 6$ hours, the distance driven changes at an average rate of **14** miles per hour.

11. The table shows the U.S. unemployment rate.

5 pts

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Rate, %	4.6	5.1	5.6	5.4	6.1	6.5	5.9	5.6	5.7	6.0	6.0

Rounded answers (2+ decimal places) are accepted.

a.) Average rate of change from 2017 to 2020 _____ (2)

b.) Average rate of change from 2023 to 2026 _____ (2)

c.) The units of these answers are... (1)

(A) percentage points per year

(B) years per percentage point

(C) percent

(D) percentage points

12. Let $f(x) = -x^2 + 4x - 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it 7 pts to fill in the table.

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) $x = 2, h = 1$ _____ (1)

c.) $x = 2, h = 0.1$ _____ (1)

d.) $x = 2, h = 0.01$ _____ (1)

e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

13. Let $f(x) = \frac{1}{x}$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ _____ (3)

b.) $f'(1) =$ _____ (2)

c.) Slope of the tangent line to f at $x = 2$ _____ (2)

d.) What does $f'(1) = -1$ mean? (1)

(A) The slope of the secant line from $x = 1$ to $x = 1 + 1$ is -1 .

(B) The value of f at $x = 1$ is -1 .

(C) The instantaneous rate of change of f at $x = 1$ is -1 — the slope of the tangent line there.

(D) The average rate of change of f from $x = 0$ to $x = 1$ is -1 .

14. (a) Suppose $f(1) = -3$ and $f'(1) = -1$. Write the equation of the tangent line to f at $x = 1$. 7 pts

(b) The tangent line to the graph of g at $x = -1$ is $y = -3x - 3$. Find $g(-1)$ and $g'(-1)$.

a.) (a) Tangent line _____ (3)

b.) (b) $g(-1)$ _____ (2)

c.) (b) $g'(-1)$ _____ (2)

15. Let $h(t) = -16t^2 + 160t$ be the height, in feet, of a rocket t seconds after launch. 6 pts

a.) $h'(t) =$ _____ (3)

b.) $h'(2)$ _____ (2)

c.) Interpret $h'(2) = 96$. (1)

(A) At $t = 2$ seconds, the height of the rocket is increasing at a rate of 96 feet per second.

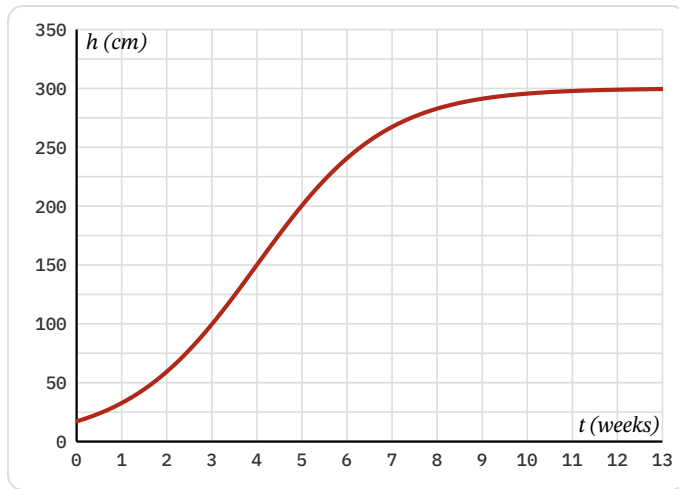
(B) At $t = 2$ seconds, the height of the rocket is 96 feet.

(C) At $t = 2$ seconds, the height of the rocket is decreasing at a rate of 96 feet per second.

(D) Over the first 2 seconds, the height of the rocket changed at an average rate of 96 feet per second.

16. A sunflower seedling is planted and its height h (in centimeters) after t weeks is graphed below. Decide whether each statement is true or false.

4 pts



- a.) T F The slopes of the tangent lines get steeper up to $t = 7$, and then the slopes start to flatten out. (1)
- b.) T F The average rate of change over $[0, 10]$ is greater than the instantaneous rate of change at $t = 3$. (1)
- c.) T F The seedling grows faster (instantaneous rate) at week 2 than at week 8. (1)
- d.) T F The average rate of change over $[10, 12]$ is less than over $[1, 3]$. (1)